



Lecture 9: Rocky Planets I

Earth & Venus

Planetary Systems

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Learning Objectives

By the end of this lecture, you will be able to:

- Use Earth's interior, surface, ocean and climate as the reference case
- Compare Venus to Earth in mass, radius, atmosphere and rotation
- Explain why a saturated water atmosphere has a maximum thermal flux
- Derive the Simpson-Nakajima runaway greenhouse limit
- Read the D/H ratio as evidence for past water on Venus

Recap: Lectures 1–8

The two innermost large rocky planets share similar bulk parameters but followed radically divergent evolutionary paths.

- Differentiation separates crust, mantle, and core ($C/MR^2 = 0.331$ for Earth)
- Volcanism, impact cratering, and escape shape surfaces and atmospheres
- Liquid outer core sustains a protective magnetic dynamo

Today: What caused the evolutionary divergence of Earth and Venus?

1. Earth as Reference

Earth seen from ~ 6 billion km. Voyager 1 (1990). Credit: NASA/JPL-Caltech

Earth at a Glance

Parameter	Value	Note
Mass	$5.972 \times 10^{24} \text{ kg}$	defines $M_{\oplus}$
Radius (mean)	6371 km	defines $R_{\oplus}$
Mean density	5515 kg m^{-3}	uncompressed $\sim 4030 \text{ kg m}^{-3}$
Semimajor axis	1.000 AU	defines AU
Bond albedo	0.30	cloud + ice + land
T_{eq}	255 K	equilibrium, no greenhouse
T_{surf}	288 K	+33 K greenhouse warming
Rotation period	23.93 h	spin axis tilt 23.4°
Magnetic field	dipolar, $\sim 25\text{--}65 \mu\text{T}$	active dynamo

Source: NASA Earth Fact Sheet (NSSDC)

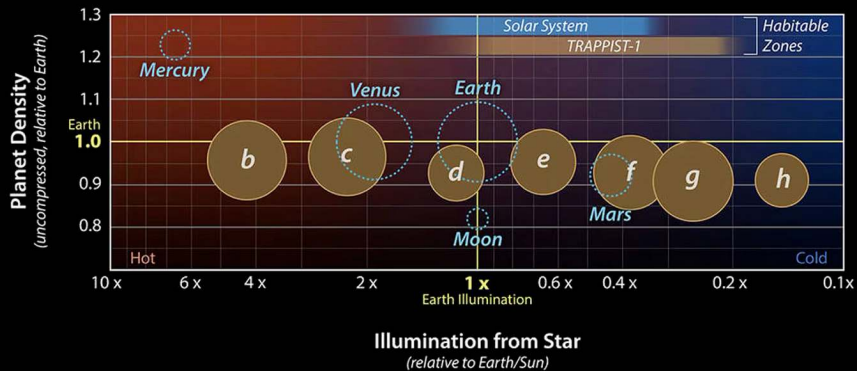
What Makes Earth Habitable?

Four ingredients, all present and stable for ~ 4 Gyr:

1. **Liquid water** on the surface, continuously
2. **Plate tectonics**: long-term carbon cycle, climate buffering
3. **Magnetic field**: shields atmosphere from solar wind erosion
4. **Stable orbit**: low eccentricity, large stabilising Moon

Identify which planetary features are necessary, which are sufficient, and which are accidental.

TRAPPIST-1/Solar System Comparison



Credit: NASA

TRAPPIST-1 vs. inner solar system, to scale. Seven Earth-sized worlds span the habitable zone around a late M dwarf, testing terrestrial evolutionary pathways. Credit: NASA/JPL-Caltech (PIA22093).

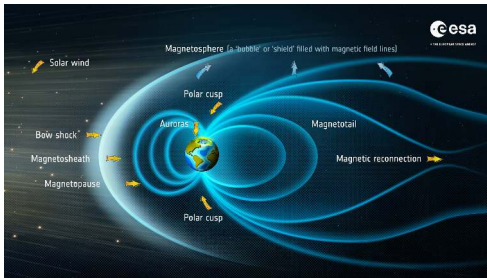
Plate Tectonics: The Engine

Earth alone among known solar system bodies operates active plate tectonics today, driving continuous volatile recycling between surface and interior.

- Lithosphere divided into ~ 15 major plates moving at cm yr^{-1} speeds
- Subduction zones recycle crust and surface volatiles on ~ 100 Myr timescales
- Closes the long-term carbon cycle by returning buried carbonate to the mantle

Earth's Magnetic Field

A vigorous geodynamo generates a protective dipolar magnetosphere that shields Earth's atmosphere from direct solar wind erosion.



Credit: ESA, CC BY-SA 3.0 IGO

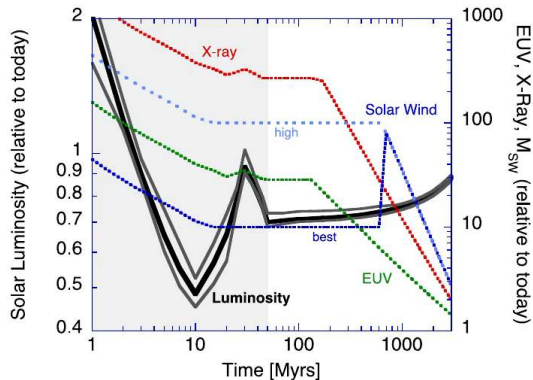
- Surface dipolar intensity of $25\text{--}65\ \mu\text{T}$ driven by core convection
- Magnetopause dayside stand-off distance at $\sim 10 R_{\oplus}$
- Deflects solar wind and suppresses ion-pickup escape of volatiles

Hydrosphere and Cryosphere

Earth possesses a stable surface liquid ocean containing over a billion cubic kilometres of water, complemented by polar cryosphere reservoirs.

- Oceans contain $1.34 \times 10^9 \text{ km}^3$ of liquid water
- Total surface water equivalent to $\approx 2700 \text{ m}$ global equivalent layer (GEL)
- Deep mantle holds $\geq 1\text{--}2$ additional ocean masses

Climate Stability and Solar Evolution

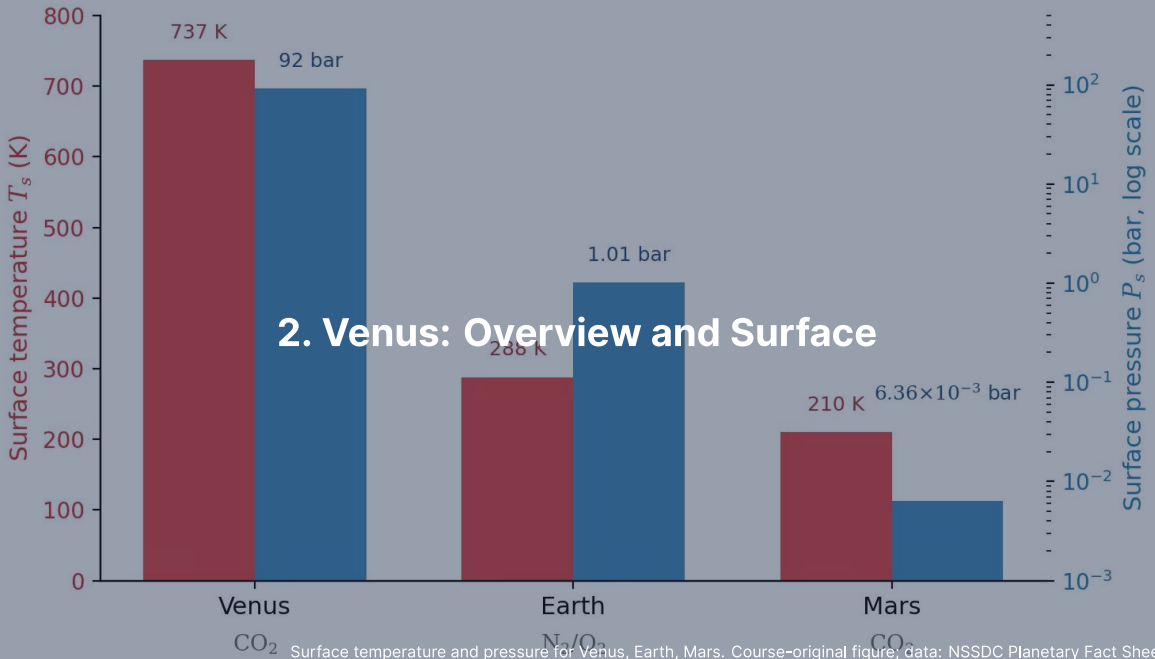


Credit: Zahnle et al. (2007)

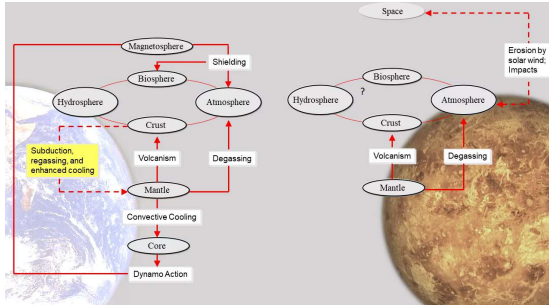
Faint young Sun problem:

- Solar luminosity increased by ~ 30% since 4.5 Ga
- Liquid water persisted at ≥ 3.8 Ga despite dimmer early Sun
- Carbonate-silicate negative feedback stabilized climate

Habitable zone inner edge migrated outward over time, sweeping past Venus.



Earth and Venus as Coupled Systems



Credit: Gillmann et al. (2022)

Both are coupled systems of:

- Interior + tectonics
- Magnetosphere or its absence
- Atmosphere + clouds
- Surface + (possible) hydrosphere

The differences between them are emergent properties of which feedback loops close.

Venus and Earth: Twin and Not

Parameter	Earth	Venus
Mass ($M_{\oplus}$)	1.000	0.815
Radius ($R_{\oplus}$)	1.000	0.949
Mean density (kg m^{-3})	5515	5243
Semimajor axis (AU)	1.000	0.723
Stellar flux ($S_{\oplus}$)	1.00	1.91
Bond albedo	0.30	0.77
T_{surf} (K)	288	737
Surface pressure (bar)	1.0	92
Atmosphere	N_2/O_2	CO_2/N_2
Rotation period (d)	+1.0	-243 (retrograde)
Magnetic field	$\sim 50 \mu\text{T}$	$< 10^{-5} \times \text{Earth}$

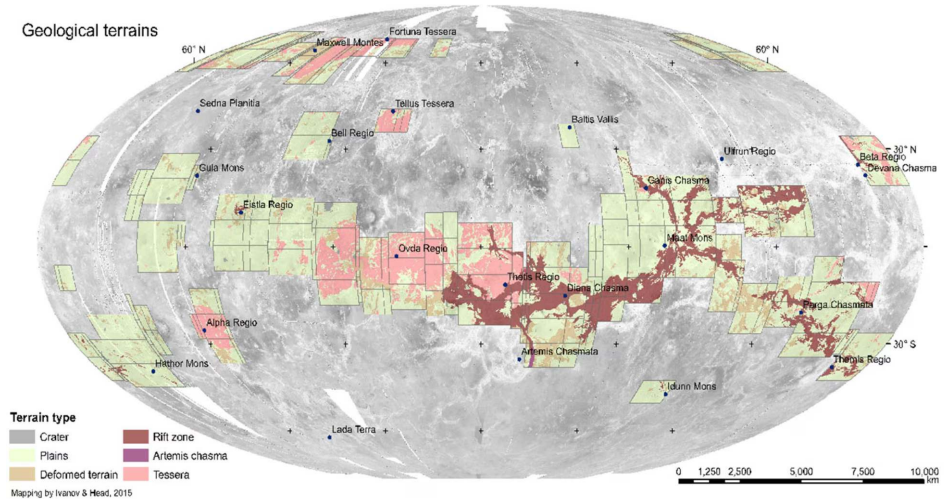
Source: NASA Venus Fact Sheet (NSSDC)

Venus: A Long Mission History

Hostile surface conditions of 92 bar and 737 K limited lander lifespans to two hours, focusing exploration on radar and orbital remote sensing.

- **Venera** (1975–1982): first surface images and XRF basaltic composition
- **Magellan** (1990–1994): global radar mapping at ~ 100 m resolution
- **Venus Express & Akatsuki** (2006 to present): atmospheric dynamics, escape rates (Venus Express) and super-rotation

Geological terrains

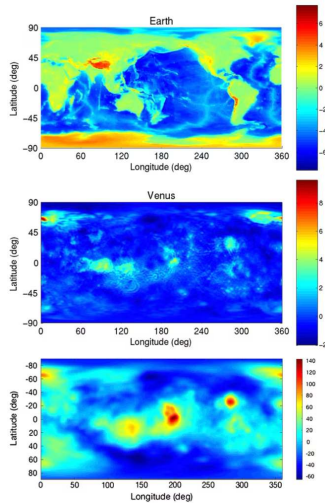


Magellan radar mosaic with terrain overlay. Volcanic plains cover ~80% of the surface alongside tessera highlands, showing no plate-tectonic boundaries. Widemann et al. (2023), SSRv 219, 56.

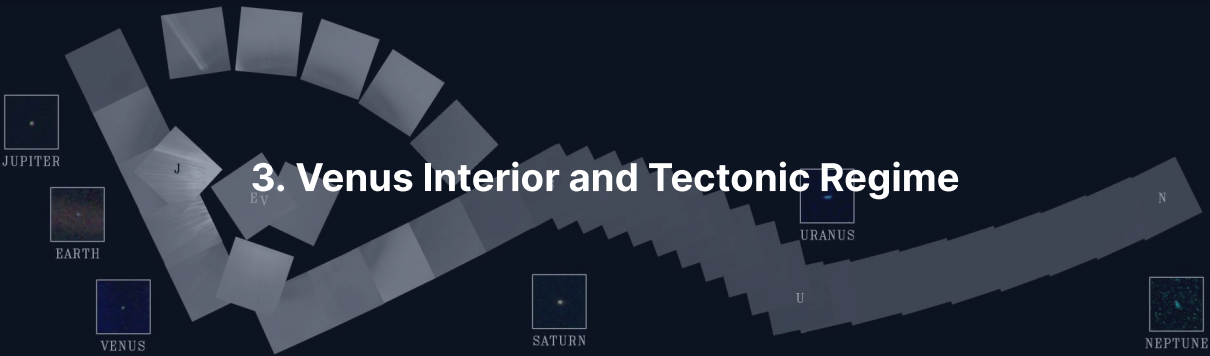
Crater Statistics: A Single Surface Age

A statistically random distribution of ~ 1000 craters yields a young global crater-retention age of 150–250 Myr across all terrains.

- ~ 1000 catalogued impact craters (≥ 2 km) show no spatial clustering
- Uniform density implies a young surface age of ~ 150–250 Myr
- Debated between catastrophic global resurfacing and steady volcanic equilibrium



Topography and geoid comparison. Unimodal hypsometry on Venus contrasts with Earth's bimodal ocean-continent distribution, ruling out Earth-style plate tectonics. Smrekar et al. (2018).



Voyager 1 Solar System family portrait (1990); each of the six planets it detected appears in a small inset box. Credit: NASA/JPL-Caltech

Venus' Interior: Inferred, Not Measured

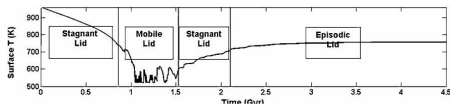
Without landed seismology or precession measurements, Venus' internal structure is modeled by Earth analogy rather than directly measured.

- Moment of inertia factor $C/MR^2 \approx 0.33$ is assumed, not measured
- Inferred metallic core radius $R_c \approx 3000\text{--}3500\text{ km}$ ($\sim 0.55 R_V$)
- Mean density of 5243 kg m^{-3} requires a substantial iron core

Venus Tectonic Evolution: A Model Trajectory

Surface temperature evolution from one numerical model: successive intervals of

- **Stagnant lid**
- **Mobile lid**
- **Stagnant lid**
- **Episodic lid**



Credit: Smrekar et al. (2018)

Conceptual alternatives debated for Venus today: **steady stagnant** (localised mantle plumes) vs. **episodic catastrophic overturn**. Either way, no Earth-style plate tectonics.

The Missing Magnetic Field

Venus lacks an intrinsic dipole magnetic field, leaving its upper atmosphere exposed to direct solar wind interaction and erosion.

- Internal magnetic field measured at $< 10^{-5}$ times Earth's field
- Stagnant-lid insulation suppresses core heat loss, shutting down thermal convection
- Unshielded atmosphere enables solar-wind induced escape and ion pickup

Why Stagnant Lid?

A hot, dry surface produces a strong lithosphere that resists subduction, locking Venus into an immobile stagnant-lid convective regime.

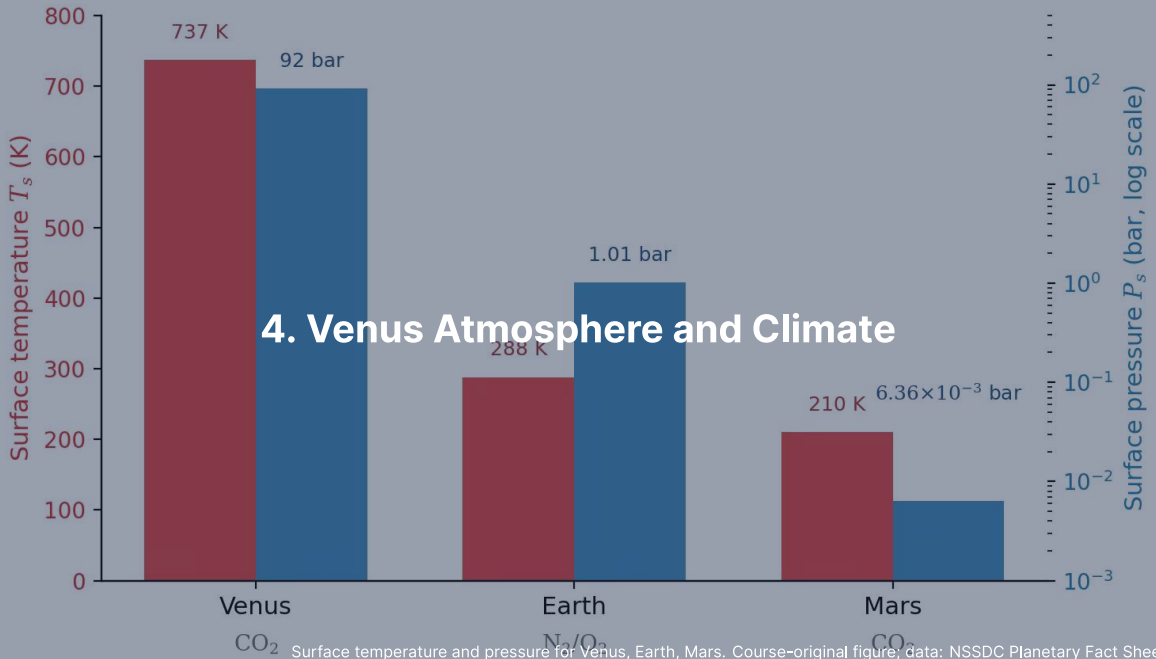
- Plate subduction requires a hydrated, mechanically weakened lithosphere
- High surface temperatures (737 K) promote ductile flow rather than brittle faulting
- No subduction return leg breaks the long-term carbon cycle permanently

Source: Smrekar et al. (2018); Gillmann et al. (2022)

Is Venus Volcanically Active Today?

Radar and orbital observations demonstrate that Venus remains volcanically active today, driven by internal heat loss through mantle plumes.

- Magellan radar reanalysis reveals a $\sim 2 \text{ km}^2$ vent change on Maat Mons (1991–1992)
- Venus Express thermal emissivity anomalies indicate lava flows younger than 250 kyr
- Decadal fluctuations in atmospheric SO_2 suggest episodic volcanic outgassing



Venus Atmosphere: Structure

Venus hosts a massive 92-bar carbon dioxide atmosphere enveloped in global sulfuric acid clouds without a tropopause cold trap.

- Composition: 96.5% CO₂, 3.5% N₂ at 737 K and 92 bar surface pressure
- Global cloud deck at 48–70 km composed of ~ 75% sulfuric acid droplets
- Lower atmosphere follows a dry adiabat with lapse rate $\Gamma \approx 8 \text{ K km}^{-1}$; scale height $H \approx 15.9 \text{ km}$
- $^{40}\text{Ar}/^{36}\text{Ar} \approx 1.1$ (Earth ~ 300): Venus has degassed only a fraction of its interior

Atmospheric Super-Rotation

Venus exhibits atmospheric super-rotation, where high-altitude cloud winds encircle the planet sixty times faster than the underlying solid surface.

- Cloud-top winds of $\sim 100 \text{ m s}^{-1}$ circle the planet in ~ 4 days
- Solid-body retrograde rotation takes 243 days (60× angular velocity ratio)
- Maintained by wave transport and atmospheric thermal tides

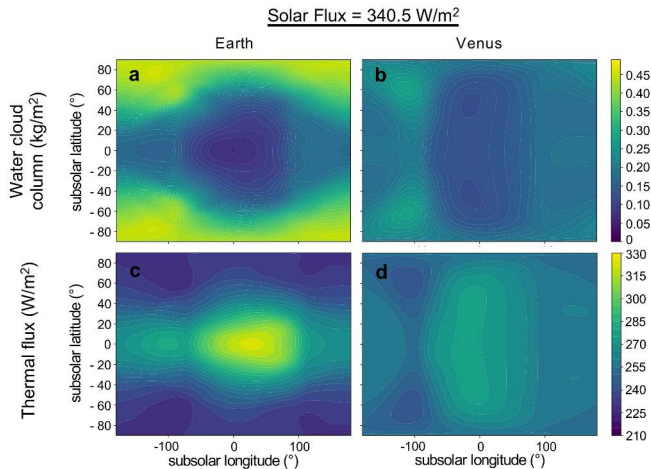
Venus' Greenhouse: T_{surf} from T_{eq}

Despite absorbing less solar energy than Earth due to high albedo, Venus experiences extreme greenhouse warming from its massive carbon dioxide atmosphere.

$$T_{\text{eq}} = \left[\frac{S(1 - A)}{4\sigma} \right]^{1/4} \approx 227 \text{ K}, \quad T_{\text{surf}} = 737 \text{ K}$$

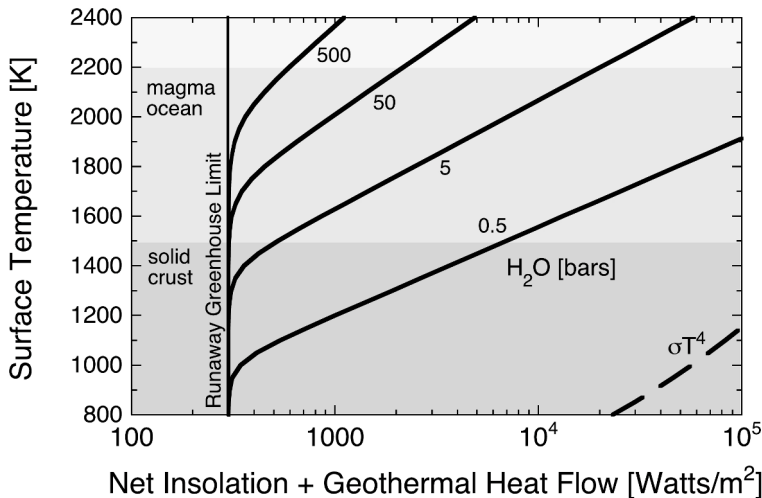
- Bond albedo $A = 0.77$ reflects most sunlight, yielding $T_{\text{eq}} = 227 \text{ K}$
- One-layer greenhouse $T_s = 2^{1/4} T_{\text{eq}} \approx 270 \text{ K}$ falls far short
- Greenhouse warming $\Delta T = +510 \text{ K}$ dwarfs Earth's $+33 \text{ K}$
- Driven by 92 bar of CO_2 , with SO_2 , H_2O and the clouds closing the spectral windows

Where the Night-Side Clouds Form



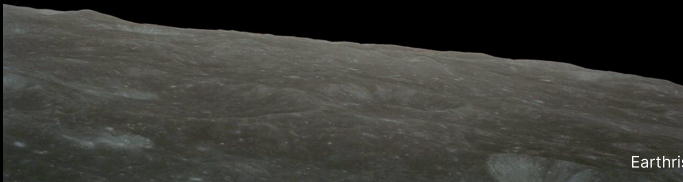
Credit: Turbet et al. (2021), Fig. 2a-d

- Hot and steamy Earth (a, c) and Venus (b, d) at the same flux, 340.5 W/m^2
- Clouds gather on the night-side in both, although Venus turns 243 times slower: the pattern does not need slow rotation
- Night-side clouds trap thermal radiation, so the emission maps are anticorrelated with the cloud maps



Thermal radiation limit and runaway greenhouse threshold. A saturated atmosphere has a maximum OLR of $\sim 280\text{--}300 \text{ W m}^{-2}$, beyond which surface water vaporises completely. Zahnle et al. (2007).

Break



5. Blackboard Derivation: Simpson-Nakajima Limit



Voyager 1 Solar System family portrait (1990); each of the six planets it detected appears in a small inset box. Credit: NASA/JPL-Caltech

Setup: Saturated Atmosphere, Greybody

Goal: show that a water-saturated atmosphere has a maximum OLR independent of surface temperature.

Assumptions:

- Atmosphere saturated in water vapour everywhere
- Single grey IR opacity κ , units $\text{m}^2 \text{kg}^{-1}$
- Hydrostatic equilibrium, surface gravity g
- Photosphere: where IR optical depth $\tau \sim 1$

Clausius-Clapeyron sets the water vapour partial pressure:

$$\frac{dp_{\text{sat}}}{dT} = \frac{L p_{\text{sat}}}{R_v T^2}$$

with $L \approx 2.5 \times 10^6 \text{ J kg}^{-1}$, $R_v = 461 \text{ J kg}^{-1} \text{ K}^{-1}$.

Result: The OLR Asymptote

Optical depth unity fixes the photosphere pressure, saturation fixes its temperature (board):

$$p_{\text{phot}} = \frac{g}{\kappa} = p_{\text{sat}}(T_{\text{phot}}) \Rightarrow T_{\text{phot}} \approx 260 \text{ K}$$

$$F_{\text{OLR}}^{\text{max}} \approx \sigma T_{\text{phot}}^4 \approx 5.67 \times 10^{-8} \times (260)^4 \approx 260 \text{ W m}^{-2}$$

Refined line-by-line calculations: $F_{\text{OLR}}^{\text{max}} \approx 280\text{--}310 \text{ W m}^{-2}$ (Kasting 1988; Goldblatt et al. 2013).

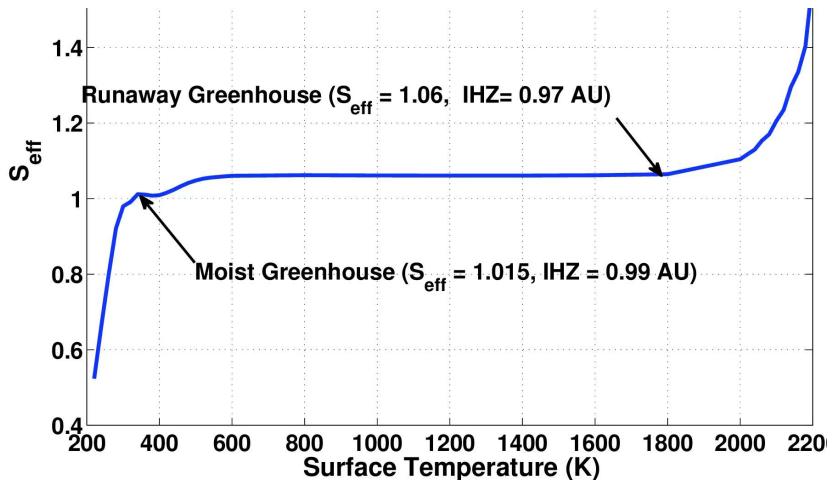
As T_s rises, the photosphere migrates upward but its temperature stays clamped: thermal flux saturates.

Against the Limit: Earth Below, Venus Above

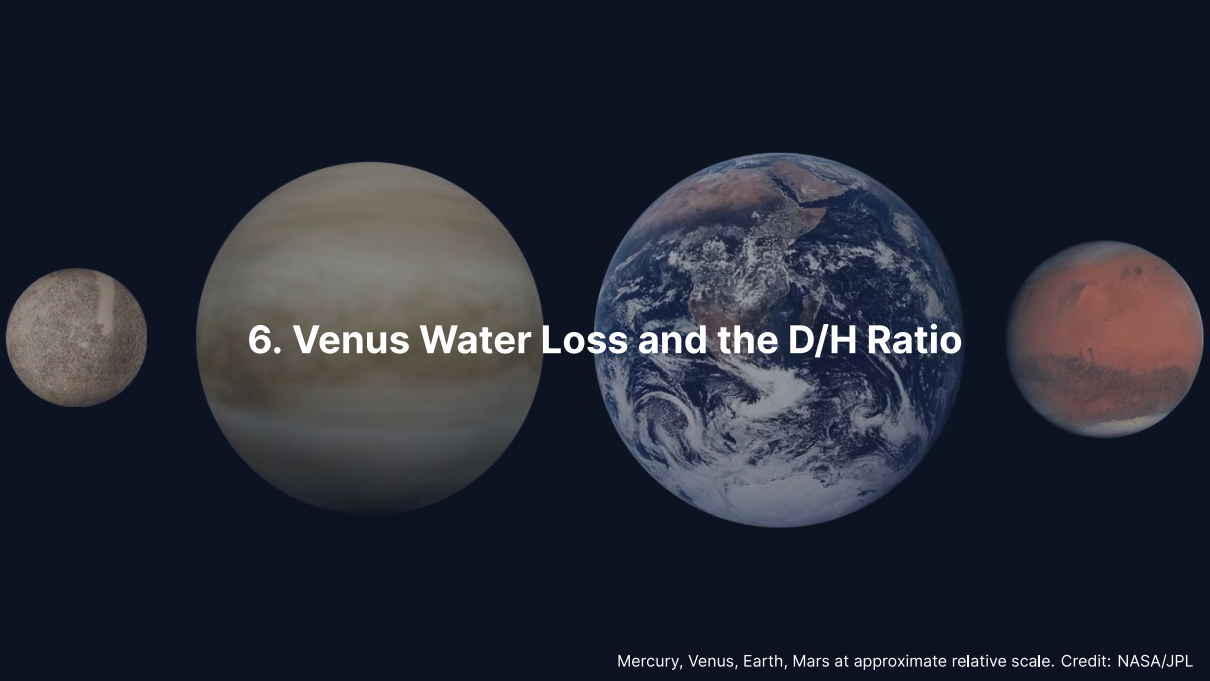
$$F_{\text{abs}} = \frac{S_{\odot}}{4a^2} (1 - A)$$

- Earth ($a = 1$ AU, $A = 0.30$): $F_{\text{abs}} \approx 240 \text{ W m}^{-2}$, below $F_{\text{OLR}}^{\text{max}}$
- Early Venus ($a = 0.723$ AU, Earth-like A): $F_{\text{abs}} \approx 460 \text{ W m}^{-2}$, far above it
- Present Venus ($A = 0.77$) absorbs only $\sim 150 \text{ W m}^{-2}$: the albedo is the product of the runaway, not its cause

Setting $F_{\text{abs}} = F_{\text{OLR}}^{\text{max}}$ gives the inner edge of the habitable zone,
 $a_{\text{IHZ}} \approx 0.95\text{--}0.99$ AU.



Inner habitable zone calculation. The runaway greenhouse limits the inner edge to $S_{\text{eff}} = 1.06$ (0.97 AU around the present Sun), placing Venus ($1.91 S_{\oplus}$) well inside. Kopparapu et al. (2013).

A horizontal arrangement of four celestial bodies against a dark blue background. From left to right: Mercury, a small greyish-brown sphere; Venus, a large pale yellowish-brown sphere; Earth, a large blue and white sphere showing continents and clouds; and Mars, a medium-sized reddish-brown sphere. The text '6. Venus Water Loss and the D/H Ratio' is centered over the Venus and Earth spheres.

6. Venus Water Loss and the D/H Ratio

Mercury, Venus, Earth, Mars at approximate relative scale. Credit: NASA/JPL

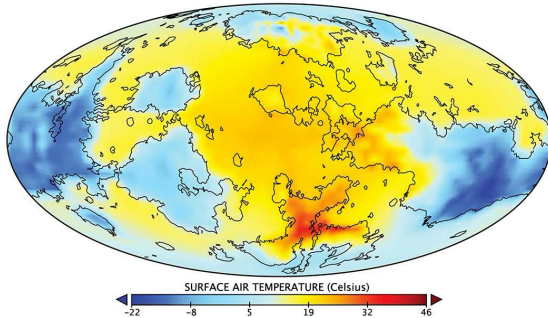
When Did Venus Lose Its Water?

Two hypotheses: early loss from a magma ocean, or late desiccation of a temperate ocean.

- **Early loss (Hamano Type II):** the steam atmosphere never condenses; water escapes during the magma-ocean phase
- **Late loss (Way et al. 2016):** a temperate surface persists for Gyr until solar brightening triggers the runaway
- The timing needs noble-gas isotopes and surface mineralogy from the next missions

Was Venus Ever Habitable? The Late-Loss Scenario

(a) Paleo Venus Surface Air Temperature (2.9Gya solar spectrum)



Credit: Way et al. (2016)

- GCM snapshot at 2.9 Gya: surface T ranges from -22°C polar to $+46^{\circ}\text{C}$ equatorial
- Cloud feedback from slow rotation reflects flux back to space
- Across the Way et al. (2016) ensemble, global mean $T < 290\text{ K}$ can persist for $\sim 1\text{ Gyr}$
- A habitable Venus is dynamically possible, not required

The Loss Mechanism: Photolysis + Escape

Massive water loss proceeds sequentially via stratospheric photolysis, rapid hydrogen escape to space, and residual oxygen oxidation or ion pickup.

- UV photolysis ($\text{H}_2\text{O} + h\nu \rightarrow \text{H} + \text{OH}$) dissociates water in the upper atmosphere
- Light hydrogen escapes efficiently through thermal Jeans and hydrodynamic escape
- Oxygen is absorbed by crustal oxidation or stripped by solar-wind ion pickup

The D/H Ratio: Evidence for Lost Water

Venus' atmospheric D/H is $\sim 150\times$ Earth's: the record of massive past hydrogen escape.

Body	D/H ($\times 10^{-4}$)
Protosolar (H_2)	~ 0.20
Earth (oceans)	1.56
Comets (Halley, 67P)	1.5–5
Venus	~ 240
Mars	~ 8

Source: Donahue et al. (1982); de Bergh et al. (1991)

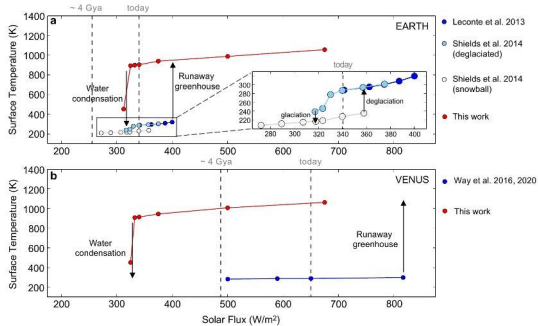
- Light 1H escapes more easily than 2H
- Rayleigh distillation:
 $R/R_0 = f^{\alpha-1}$
- Diffusion-limited ($\alpha = 0.5$):
 $f \approx 150^{-2}$, initial water ~ 450 m GEL; the Jeans limit ($\alpha = 0.71$) gives an unphysical > 100 oceans, so 450 m is a lower bound

Bounding the Initial Water Inventory

Rayleigh distillation models translate the measured deuterium enrichment into a firm lower bound on Venus' initial water inventory.

- Rayleigh distillation relation: $R/R_0 = f^{\alpha-1}$ with fractionation factor α
- Diffusion limit ($\alpha \approx 0.5$) yields remaining water fraction
 $f \approx 150^{-2} \approx 4.4 \times 10^{-5}$
- Current 2 cm GEL column implies initial water equivalent layer $d_{\text{init}} \geq 450$ m GEL

Bistability and Hysteresis Loop



Credit: Turbet et al. (2021), Fig. 4

For some stellar fluxes, two stable states coexist:

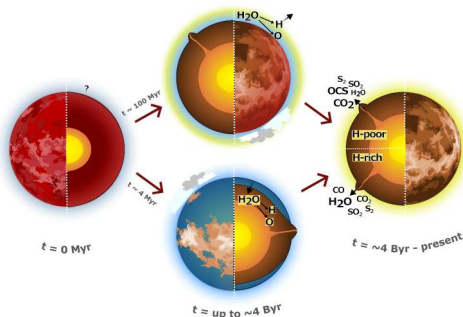
- Wet: ocean + thin atmosphere
- Dry: hot, thick CO₂ atmosphere

Crossing the threshold is one-way. The dry state has no thermodynamic path back.

Constraining the Trajectory

Two end-member pathways for Venus' water history:

- **Top branch (dry):** no surface ocean; H lost early; mantle dry today
- **Bottom branch (wet):** ocean condenses, late loss; mantle still wet today
- Constantinou et al. (2024) argue current photochemistry favours the dry/early-loss branch (volcanic $\text{H}_2\text{O} \lesssim 6\%$)
- DAVINCI noble-gas measurements will discriminate



Credit: Constantinou et al. (2024)

The image shows four planets from the solar system arranged horizontally against a dark blue background. From left to right, they are Mercury (a small, grey, cratered sphere), Venus (a large, pale yellowish-brown sphere with faint cloud bands), Earth (a large sphere with blue oceans, white clouds, and brown/green landmasses), and Mars (a medium-sized sphere with a reddish-orange surface and some darker patches). The text '7. Why Did Earth and Venus Diverge?' is centered over the Venus and Earth planets in a white, bold, sans-serif font.

7. Why Did Earth and Venus Diverge?

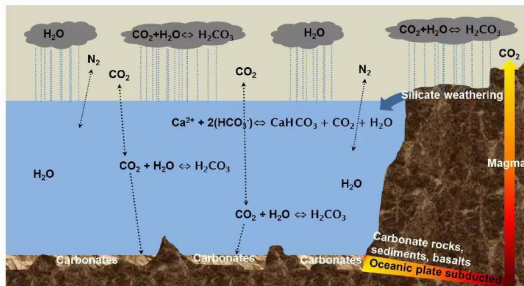
Mercury, Venus, Earth, Mars at approximate relative scale. Credit: NASA/JPL

Carbonate-Silicate Cycle: Failure Mode

Without surface liquid water and subduction, Venus lacks weathering sinks and mantle return legs, causing volcanic carbon dioxide to build indefinitely.

- Silicate weathering sink collapses entirely in the absence of liquid surface water
- Stagnant-lid regime prevents subduction return of carbonates to the mantle
- Volcanic outgassing irreversibly drives atmospheric CO_2 to the observed 92 bar

The Earth Carbonate-Silicate Schematic



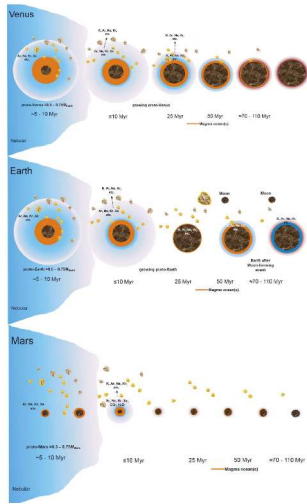
Credit: Lammer et al. (2018)

Earth's closed carbon loop:

- Volcanic outgassing (mantle to atmosphere)
- Silicate weathering on continents (atmosphere to ocean, T -dependent sink)
- Carbonate sedimentation + subduction (ocean to mantle return leg)

Venus today has no liquid-water sink and no subduction return: both halves of the loop are broken.

Initial Volatile Inventories



Credit: Lammer et al. (2018)

- Comparative accretion of Venus, Earth, Mars
- Initial water + carbon inventories overlap substantially
- Divergence is not from initial conditions alone
- Early evolution decides which branch each planet takes

Why the Twins Diverged: Habitability Implications

Coupled factors drove Earth and Venus apart:

1. **Solar flux:** $1.91\times$ Earth's insolation, above the runaway threshold
2. **Geophysics:** no intrinsic dynamo, slow retrograde rotation
3. **Tectonics:** a stagnant lid removes the subduction return leg of the carbon cycle

Venus defines the inner edge of the habitable zone: coupled interior-atmosphere feedbacks decide whether a terrestrial planet keeps its water.



8. Recent Advances and Future Missions



VERITAS, DAVINCI and EnVision over an Akatsuki UV view of Venus. Credit: Widemann et al. (2023)



The 2030s Venus fleet: DAVINCI (NASA), VERITAS (NASA), and EnVision (ESA). DAVINCI's descent probe mass spectrometer will measure atmospheric noble gases to distinguish early against late water loss. Widemann et al. (2023).

Summary

1. **Bulk twin:** same size and density, but ~ 500 K and 92 bar apart
2. **Runaway limit:** the Simpson-Nakajima threshold ($280\text{--}310 \text{ W m}^{-2}$) sets the inner habitable zone
3. **D/H evidence:** ~ 150× Earth's ratio records massive water loss
4. **Broken cycle:** no surface water stops the weathering sink; a stagnant lid removes the subduction return leg

Next: Lecture 10 Mercury and Mars.



Questions?

Next: Lecture 10. Rocky Planets II: Mercury & Mars

Lecture notes: formingworlds.github.io/IntroductionToPlanetaryScience

